

Offline, once: metric mapping + instance registration

Multiview capture → COLMAP + metric alignment → 3D Gaussian Splatting → SAM + cross-view association → Persistent IDs + one-time naming

Wearer gaze → world fixations

Camera localization
ArUco landmarks

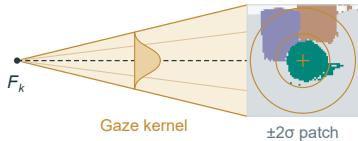
World-space gaze
Pose + rendered depth

Fixation events
Spatial + temporal clustering



3DGS-based instance disambiguation

Visible surfaces within gaze uncertainty



Visible mass → **Size correction** → **Rank + gate**
 $q_k = m_k / W$ → q_k / C_k → **ID / no binding**

Visibility · uncertainty · target angular size

Shared 3DGS instance map

One metric frame for user and robot



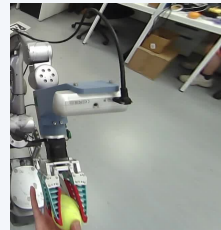
Pose-conditioned depth + opacity

Persistent IDs + metric locations

Robot execution

Grounded target → action

Map-frame navigation
Target re-observation



Grasp & deliver

Execute commands
move · pick · place

Speech → grounded commands

Streaming ASR
Word timestamps

LLM: actions + slots
"Bring me that ball."

Word-fixation binding

gaze word ——— **ball_R** ———
word ——— "that" ———

Compile + dispatch
Action + ID + world parameters
On successful binding